



AIMS

African Institute for
Mathematical Sciences
SENEGAL

Project Report Real LLM Deployment and Standard Industrial Methodologies

Wolof Educational Assistant — LoRA Fine-tuned Qwen3-0.6B

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GitHub:	https://github.com/mairamesniang12/aims-gaai-exam-wolof-assistant
HF Model:	https://huggingface.co/niangmariame513/wolof-assistant-qwen3
HF Space:	https://huggingface.co/spaces/niangmariame513/wolof-assistant-demo

1. Problem Definition and Use Case

1.1 Motivation

Wolof is the most widely spoken language in Senegal, used daily by over 10 million people. Despite its importance, Wolof remains a low-resource language in the context of modern NLP and generative AI: it is not natively supported by major ASR systems such as Whisper, and very few instruction-tuned language models exist for it. This project addresses this gap by building a small but functional instruction-following assistant fine-tuned on Wolof data, deployable on consumer hardware and accessible via a public web interface.

1.2 Use Case

The target use case is a **Wolof educational assistant** that can:

- Answer basic questions in Wolof
- Assist with simple language learning tasks
- Demonstrate that low-resource African languages can be incorporated into modern LLM fine-tuning pipelines

1.3 Base Model

We selected **Qwen/Qwen3-0.6B** as the base model because it fits in memory on a single T4 GPU with LoRA, supports chat templates with system/user/assistant roles, has strong

multilingual capabilities despite its small size, and is publicly available on Hugging Face Hub.

2. Data Methodology

2.1 Data Sources

We used three separated data sources as required by the exam methodology:

Source	Type	Train	Val	Eval
AYA (CohereLabs)	General language base	450	25	25
Soynade	Domain-specific Wolof	450	25	25
Synthetic	Instruction dataset	4770	265	265
Total		5670	315	315

2.2 Chat Formatting

Every example was converted to the `system/user/assistant` chat format required for instruction fine-tuning:

```
{
  "role": "system",
  "content": "You are a helpful Wolof language assistant."
}
{
  "role": "user",
  "content": "<instruction> <input>"
}
{
  "role": "assistant",
  "content": "<output>"
}
```

2.3 Data Splits

Split	Ratio	Total Examples
Train	90%	5670
Validation	5%	315
Eval	5%	315

3. Training

3.1 Assistant-Only Label Masking

The most important technical requirement is assistant-only training with -100 label masking. The training script `train_lora_assistant_only.py` applies the following masking:

```
system tokens    -> label = -100 (ignored in loss)
user tokens      -> label = -100 (ignored in loss)
padding tokens   -> label = -100 (ignored in loss)
assistant tokens -> normal token ids (trained on)
```

The tokenization statistics confirmed: `avg_total` = 129.0 tokens, `avg_supervised` = 33.9 tokens. Only 33.9 out of 129.0 tokens per example were used in the loss computation, confirming correct masking.

3.2 LoRA Configuration

Parameter	Value
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Trainable parameters	10,092,544 (1.67%)
Hardware	Google Colab T4 GPU
Training duration	~2.5 hours

3.3 Training Results

The training loss decreased from 5.536 at step 1 to ~0.037 at the final steps over 3 epochs. The validation loss stabilized around 0.145 after epoch 2, indicating good generalization without severe overfitting.

Epoch	Train Loss	Eval Loss
0.14	5.536 → 1.886	1.556
0.56	~0.35	0.261
1.00	~0.21	0.196
1.55	~0.18	0.162
2.00	~0.09	0.145
2.26	~0.06	0.148

4. Evaluation

4.1 Automatic Metrics

Evaluation was performed on the held-out eval set (315 examples) using `evaluation.py` with the `generate` flag. Metrics reported: Exact Match, Token F1, BLEU, ROUGE-L, and perplexity.

4.2 Qualitative Examples

Type	Input	Expected	Generated
Success	<i>Jëf-jëf bu baax.</i>	Correct Wolof response	Correct
Failure	<i>Ana nga?</i>	<i>Mangi fi rekk</i>	<i>What?</i>

Analysis: The model understood “*Ana nga?*” but responded in English (“*What?*”), showing limited Wolof generation capability with only ~6300 training examples. This is a documented limitation caused by the dominance of the synthetic dataset and the base model’s limited Wolof pre-training.

5. Deployment

5.1 Hugging Face Hub

The LoRA adapter was pushed to `niangmariame513/wolof-assistant-qwen3` using `push_to_hub.py`. Files include: `adapter_config.json`, `adapter_model.safetensors`, and `tokenizer` files. Visibility: Public.

5.2 Hugging Face Space

A Gradio application is deployed at `niangmariame513/wolof-assistant-demo`. The Space loads the LoRA adapter from Hub (not local), provides example Wolof prompts, includes a visible guardrail/limitations message, and supports adjustable `max_tokens` and `temperature` parameters.

5.3 Context State Machine Improvement

The file `src/context_state_machine.py` was improved with two additions:

- A confidence score mechanism that detects low-quality context retrieval and returns a fallback message
- A keyword-based safety filter for out-of-scope requests (medical, legal, harmful content)

6. Limitations

Small dataset

~6300 examples is insufficient for robust Wolof instruction following. A production system would require 50,000+ high-quality examples.

English fallback

The model sometimes responds in English to Wolof prompts, reflecting the dominance of English in the base model's pre-training data.

Whisper ASR incompatibility

Wolof is not natively supported by Whisper, which limits the ability to build voice-based interfaces on top of this model.

Orthography inconsistency

The Soynade dataset uses non-standard Wolof orthography, which may cause spelling inconsistencies in outputs.

No safety evaluation

The model has not been evaluated for harmful outputs in a Wolof-specific cultural context.

7. Conclusion

This project demonstrated a complete industrial-style LLM deployment pipeline applied to Wolof, a major low-resource African language. Starting from three separated data sources, we fine-tuned Qwen3-0.6B with LoRA using assistant-only -100 masking, evaluated the model on a held-out eval set, and deployed a working Gradio Space on Hugging Face. The main technical contributions are:

- A reproducible 3-source data pipeline
- Correctly implemented assistant-only training with label masking
- A public LoRA adapter and deployed Space
- Honest documentation of limitations and failure cases

Future work should focus on larger Wolof datasets, better evaluation benchmarks for African languages, and integration with a Wolof-compatible ASR system.

8. Submission Checklist

Item	Status	Link
GitHub repository	Done	github.com/mairamesniang12/aims-gaai-exam-wolof-assistant
HF model repository	Done	https://huggingface.co/niangmariame513/wolof-as
HF Space (running)	Done	https://huggingface.co/spaces/niangmariame513/w
Model card completed	Done	In HF model repository
Project report	Done	This document
Individual technical notes (x5)	Done	One per group member
Assistant-only -100 masking	Done	Verified in training output
<code>context_state_machine.py</code> improved	Done	Confidence score + safety filter